

Introduction to the Internet of Things (I03031)

Date: 1st June, 2026
Course Instructor (s)
Ms. Abeer Bashir

Final Exam

Total Time (Hrs): 3
Total Marks: 100
Total Questions: 4

Roll No

Section

Student Signature

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Instructions:

1. Do not forget to write your Name and Roll Numbers.
2. Make assumptions if you think some data is missing.

CLO #1: Use understanding of IoT architecture and identify practical application areas (PLO 1, C3)

Q1: Draw and illustrate a properly labeled architecture of a Smart Healthcare IoT System for remote patient monitoring. Your diagram should clearly depict all major components involved in the IoT system. [15 marks]

- Illustrate** and label the direction of data flow throughout the system.
- Determine** the most suitable application-layer protocols used for communication between sensors-to-gateway and cloud-to-application and justify your selection.
- For the following scenarios, **select** the most appropriate application layer protocol and justify your answer:
 - Emergency alert transmission (critical data)
 - Continuous patient monitoring (non-critical data)
 - Mobile/web application access

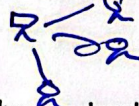
CLO # 2: Apply the knowledge of Internet Connectivity Principles, IoT Networking Technologies, Protocols, and standards (PLO1, C3)

Q2: [15+10+5 marks]

- Construct and illustrate** a properly labeled LoRa network architecture and give the following answers:
 - Classify** and explain the different LoRaWAN device classes and justify their suitability for different IoT applications.
 - Examine** any three differences between LoRa and LoRaWAN.

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- iii. Compare and evaluate LoRa, Wi-Fi, and ZigBee in terms of range, power consumption and data rate.
- b. Explain NB-IoT and **interpret** its deployment modes with the help of figures. Compare NB-IoT with LoRa in terms of coverage, data rate, and cost. **Select** and justify which technology is more suitable for any two different IoT application scenarios.
- c. Explain 6LoWPAN and **illustrate** the labeled 6LoWPAN network architecture.



CLO # 3: Explore IoT applications based on hardware, cloud storage and data analysis for industrial and societal usage (PLO 6, C4)

Q3:

[10+10 marks]

a. **Analyze** the Fog computing architecture and give the answers for each part:

- i. **Illustrate** its architecture with a neat diagram.
- ii. **Identify** and **appraise** any three benefits of Fog Computing in IoT systems.
- iii. **Identify** any three limitations of Fog Computing.
- iv. **Examine** whether fog computing is preferable when long-term data storage and extensive computational processing are required.

b. A smart city has deployed an IoT-based traffic management and emergency response system. The system collects real-time data from roadside sensors, CCTV cameras, and connected vehicles. The requirements of the system are as follows:

- Traffic signal control must react within milliseconds
- Video analytics from cameras should be processed near the data source
- Long-term traffic patterns must be stored and analyzed
- Emergency vehicle detection requires immediate response and alerting

Analyze this IoT-based traffic management system and give the following answers:

- i. **Select and justify** the most suitable computing paradigm (Edge, Fog, or Cloud) for each above mentioned requirement.
- ii. **Compare** and justify whether a single-layer architecture is sufficient or a multi-layer architecture is required for this system.

CLO # 4: Evaluate IoT solutions in terms of their environmental impact and contribution to sustainable development (PLO 7, C5)

Q4:

[20+15 marks]

- a. An IoT-based health monitoring system collects data from wearable sensors. **Formulate and evaluate** the classification of the new patient using the Gaussian Naive Bayes classifier. Dataset is given below:

A new patient has the following readings: Heart Rate = 100, Body Temperature = 38.0, SpO₂ = 94

$P(N) = 6.2 \times 10^{-6}$
 $P(C) =$

crit $\left[\begin{array}{l} m = \\ \sigma = 58.335 \end{array} \right]$ $\left[\begin{array}{l} m = 38.9 \\ \sigma = 0.13 \end{array} \right]$ $\left[\begin{array}{l} m = 90 \\ \sigma^2 = 4 \end{array} \right]$

Patient ID	Heart Rate	Body Temperature	SPO ₂	Condition
P1	78	36.8	98	Normal
P2	110	38.5	92	Critical
P3	85	37.2	96	Normal
P4	120	39.0	90	Critical
P5	90	37.5	95	Normal
P6	125	39.2	88	Critical

- b. A smart city IoT system collects environmental and traffic-related sensor data. The dataset is given below:

Traffic count	CO ₂ Level
1200 ✓	850
3500 ✓	1200
5000 ✓	1500
2800 ✓	980
6000 ✓	1700

$$a_0 = 0.46865$$

$$b_0 = 0.76789$$

$$Y = 1.2365$$

- Formulate and evaluate** the linear regression model using the Least Squares method for the given dataset.
- Determine and evaluate** the loss function using Mean Squared Error (MSE).
- Perform one full epoch of gradient descent, and update the parameters. Show all calculations clearly.
- Predict and evaluate** the CO₂ level when traffic count $x=7000$ using the updated regression model.